Collaborative Docs >

Discussion Agenda 04/24/2017 (Science-oriented Comparison)

Below is a discussion agenda / live progress report for the 10th Web Conference of the AGORA Project (Apr. 24, 12 pm PDT/9 pm CEST; **screenshot**). This telecon will particularly focus on the future project and direction of AGORA. Some items below were discussed amongst the AGORA Steering Committee members on Mar. 10, 13, and 27 in preparation of this meeting.

Contents

Check out the important items in red color below!

(last revision: 04/26/2017)

1. News and Topics for Collaboration-wide Discussion

(1) AGORA Isolated Disk Comparison Paper Published

- See **this blog link** for details.

(2) 6th Annual AGORA Workshop

- This year's annual Santa Cruz Galaxy Workshop will be at UC Santa Cruz, August 7-11, 2017. The concluding day (Friday) of the Galaxy Workshop will be the first day of the **6th AGORA Workshop**, August 11-13, 2017.

- See Section 2 below for further discussion about its planning and agenda.

(3) Future Funding and Logistics in General

- At this point, there is no immediate prospect for future funding. However, some leftover funds from UC-HIPACC are available for this year's Workshop thanks to Joel Primack. Even so, we will likely need to ask for a small registration fee from each participant.

- We are looking for future sources of support. A proposal may be submitted to the Simons Foundation after pre-consultation with Greg Bryan and Rachel Somerville, who co-chair the galaxy group at the Simons Center for Computational Astrophysics, New York.

- Any institution interested in supporting the logistics of future AGORA efforts? U. Zurich is happy to contribute financially to organize a future workshop in Zurich, e. g. in 2018 and possibly later. The level of financial support is to be determined. [comment by Mayer/Teyssier] Stanford U. is happy to host small workshops between annual ones, and potentially the annual workshop in 2018 -- but maybe not on weekends. [comment by Abel] As for the long-term needs for the Collaboration, ideally, we would need at least one postdoc with more than one third of his/her time dedicated to AGORA. This is to keep track of various progresses in the Collaboration, to organize meetings and telecons, to analyze and manage the data, etc. [thoughts by Kim/Madau/Primack]

2. Science Paper(s) to Focus In This Year's Workshop

(1) Base Philosophy - Proposition

• Before and during this year's Workshop, let us focus on science-oriented paper(s), rather than technical papers (e.g., Paper "DM-Highres" or Paper "NoSubgrid"). We will design this year's Workshop to showcase how to launch smaller, science-oriented comparisons (i.e., 3-4 codes rather than the entire 9 codes).

• We will choose 1-2 "fast" science topics that could garner the most interest from the community, from observers, and from prospective paper leaders. Interested postdocs/students will lead this project in the next 1-2 years until the publication of the results. The key is to encourage young students and postdocs to take the leaderships and ownerships, driven by their own science interests (rather than by a top-down directive from the Collaboration). It is important to help them to quickly acquire desired outcomes (i.e., publishable results).

• The topics should not carry too high an entry barrier in terms of numerical resolution or subgrid physics requirements. [thoughts by Kim/Primack] The topics should be well-posed and not completely dominated by subgrid physics. [thoughts by Mayer] Comparison of cosmological zoom-in runs (10^{11} or 10^{12} Ms halos) will be most interesting scientifically. See the next sections for example topics, and how to submit yours.

(2) Selected Science Topics (as of Apr. 24, 2017)

*--> **Below are already outdated; see the most recent "Discussion Agenda" for updated information about these Paper Groups... (comment on Jun. 28, 2017)***

The summary below is based on today's telecon, and the post-telecon discussions among the group leaders.

<table>
<tr><th></th><th>Paper "CGM"</th><th>Paper "Clumps"</th></tr>
<tr><td>GOALS</td><td>• Compare the CGM absorption lines in simulated star-forming galaxies.</td><td>• Compare the off-center star-forming clumps in simulated star-forming galaxies.</td></tr>
<tr><td>LEADERS</td><td>• Cameron Hummels (Caltech)
• Sijing Shen (U. Oslo)</td><td>• Daniel Ceverino (U. Heidelberg)
• Alessandro Lupi (IAP)</td></tr>
<tr><td>PARTICIPANTS</td><td>• Interested code representatives:
<table>
<tr><th>Code</th><th>Contacts</th></tr>
<tr><td>ART-I</td><td>Mandelker, Primack, Roca-Fabrega, Strawn</td></tr>
<tr><td>ENZO</td><td>Hummels, Kim, O'Shea, Wise</td></tr>
<tr><td>GADGET</td><td>Nagamine</td></tr>
<tr><td>GASOLINE</td><td>Shen</td></tr>
<tr><td>GIZMO</td><td>(Hummels)</td></tr>
<tr><td>[to be added]</td><td></td></tr>
</table>
(10^{12} Ms ellipsoidal ICs: Hahn, Simpson / analysis
help: Chakrabarti, Fumagalli, Hummels, Kim, Roca-Fabrega, Smith, Turk)

• Working Groups engaged:
- III (common cosmo ICs)
- IV (common analysis)
- V (subgrid physics)
- IX (Characteristics of Cosmo. Galaxies)
- X (Outflows)</td><td>• Interested code representatives:
<table>
<tr><th>Code</th><th>Contacts</th></tr>
<tr><td>ART-I</td><td>Ceverino, Mandelker, Primack</td></tr>
<tr><td>GADGET</td><td>Nagamine</td></tr>
<tr><td>GASOLINE</td><td>Mayer, Wadsley</td></tr>
<tr><td>GIZMO</td><td>Lupi</td></tr>
<tr><td>[to be added]</td><td></td></tr>
</table>
(analysis help: Chakrabarti, Kim, Turk)

• Working Groups engaged:
- IV (common analysis)
- XI (High-redshift Galaxies)</td></tr>
<tr><td>SCIENCE CASE</td><td>• We will compare the run with observed absorption lines data from e.g., KBSS at high redshift (Rudie et al. 2012, Steidel et al. 2014), or COS-Halos at low redshift (webpage, Tumlinson et al. 2013). Observers will certainly be interested.

• We may later add MHD for comparison.</td><td>• Observationally, most star-forming galaxies have off-center clumps (e.g., Guo et al. 2015).
• Can we resolve recent conflicting results by ART-I, GASOLINE/ChaNGa, GIZMO?
• Numerical issues such as viscosity, dissipation, non-conservation of angular momentum could be discussed. The topic is still impacted by numerical implementation of</td></tr>
</table>

<table>
<tr><td></td><td></td><td>the hydro as much as subgrid physics.
• A related topic is secular disk instabilities. How do different hydro codes (and their gravity solvers) generate bulges of spiral galaxies and impact galactic morphology?</td></tr>
<tr><td>PROGRESS MADE OR BEING MADE</td>
<td>• Pre-production discussion/to-do's:
<table>
<tr><th>Item</th><th>To-do</th></tr>
<tr><td>Initial condition</td><td>10^{12} Ms halo
(ellipsoidal IC to be delivered by Hahn and Simpson)</td></tr>
<tr><td>Subgrid physics</td><td>The stellar fbck prescription of each code group's choosing needs to be first calibrated across codes to produce similar properties
(e.g., stellar masses), and to eject metals into CGM.</td></tr>
<tr><td>Target redshift</td><td>z=2.5 (initial target),
then
z=0 (secondary target)</td></tr>
<tr><td>Output strategy</td><td>See Section 4-(2) below.</td></tr>
<tr><td>Analysis tools</td><td>The yt-<u>Trident</u> pipeline has been field-tested for a few codes (ART-I/ENZO/GIZMO), but needs to be fixed for some SPH codes (Cameron; issue reported by Dave/Somerville in GADGET) possibly via <u>this effort</u> in yt.</td></tr>
<tr><td>Obs. counterpart</td><td>e.g., KBSS at high-z, COS-Halos at low-z
(to be determined)</td></tr>
</table>
• Upcoming group-level telecon:
<table>
<tr><td>Date</td><td>early July, 2017
(to be decided after all the code reps. are identified)</td></tr>
<tr><td>First homework</td><td>(to be decided soon)</td></tr>
</table>
</td>
<td>• Pre-production discussion/to-do's:
<table>
<tr><th>Item</th><th>To-do</th></tr>
<tr><td>Initial condition</td><td>10^{13} Ms halo
(ellipsoidal IC to be retrieved from Madau's group)</td></tr>
<tr><td>Subgrid physics</td><td>The stellar fbck prescription of each code group's choosing needs to be first calibrated across codes to produce similar properties (e.g., stellar masses), and to overcome the overcooling problem in this massive halo. More effective fbck (including AGN) may need to be considered.</td></tr>
<tr><td>Target redshift</td><td>z=2-6
(to be determined)</td></tr>
<tr><td>Output strategy</td><td>See Section 4-(2) below.</td></tr>
<tr><td>Analysis tools</td><td>We may begin from clump analysis routines in <u>agora-analysis-script</u> used for our disk comparison paper (formerly <u>Paper "4"</u>) and build upon it.</td></tr>
<tr><td>Obs. counterpart</td><td>e.g., Guo et al.
(to be determined)</td></tr>
</table>
• Upcoming group-level telecon:
<table>
<tr><td>Date</td><td>June 16, 2017</td></tr>
<tr><td>First homework</td><td>(to be decided soon)</td></tr>
</table>
</td></tr>
<tr><td>OTHER LINKS</td>
<td>• <u>Trident download and documentation</u>
• <u>Trident method paper (Hummels et al.)</u>
• <u>WG III Workspace (AGORA Cosmo. ICs)</u>
• <u>WG IX Workspace (Char. of Cosmo. Gals)</u>
• <u>WG X Workspace (Outflows)</u></td>
<td>• <u>WG XI Workspace (High-z Galaxies)</u></td></tr>
</table>

(3) Science Topics To Be Considered *Later*

--> Below are already outdated; see the most recent "Discussion Agenda" for updated information about these Paper Groups... (comment on Jun. 28, 2017)

	Paper "Metallicity"
GOALS	• Connect galaxy evolution and LIGO observations.

<table>
<tr><td>LEADERS</td><td>(• Sukanya Chakrabarti (Rochester IT) - proposal submitter)</td></tr>
<tr><td>PARTICIPANTS</td><td>• Interested code representatives:
<table>
<tr><th>Code</th><th>Contacts</th></tr>
<tr><td>GADGET/GIZMO</td><td>Chakrabarti, Lewis</td></tr>
<tr><td>GEAR</td><td>(Revaz)</td></tr>
<tr><td>[to be added]</td><td></td></tr>
</table>
(Butler may be interested in running tests using an isolated disk IC with ENZO, in connection with Paper "Disk-II"; see Section 3 below.)

• Working Groups engaged:
- IX (Characteristics of Cosmological Galaxies),
- XII (Interstellar Medium)</td></tr>
<tr><td>SCIENCE CASE</td><td>• Belczynski et al. 2016 recently showed that the LIGO detection of GW150914 could be explained by the merger of massive, low-metallicity binary stars. A question that remains is what kind of galaxies would these progenitors be typically found in?
• Initial work on understanding the host galaxies of LIGO events was done by O'Shaughnessy et al. 2016, showing that dwarf galaxies may contribute significantly to the LIGO signal, due to their low metallicity. Sufficiently low metallicity environments may also be found in the outer disks of spiral galaxies, which have yet to be studied.
• The AGORA infrastructure is ideal for testing the effects of feedback on metallicity. It is already clear from Kim et al. 2016 that various solvers have some effect on the metallicity. Different kinds of feedback models are also likely to affect the metallicity in galaxy disks, and given the strong dependence on metallicity on LIGO detections, this in turn is likely to determine the most probable host galaxies for BH-BH mergers.
• Regarding connecting LIGO to galaxy evolution, there's a possible connection using chemical evolution within galaxies (e.g., Cote et al. 2017). Interesting opportunities may exist to connect this to AGORA, given that some of the most interesting questions about NS-NS mergers come from the behaviors of ultra-faint dwarf galaxies (e.g., the recent Reticulum II work by Ji et al. 2016 and others). [thoughts by O'Shea]
• Metal diffusion : comparison between different codes/methods [thoughts by Revaz]</td></tr>
<tr><td>PROGRESS MADE OR BEING MADE</td><td>• Pre-production discussion/to-do's:
<table>
<tr><th>Item</th><th>To-do</th></tr>
<tr><td>Initial condition</td><td>10^{12} Ms halo
(we will piggyback on the planned CGM run above and evolve them further down to z=0; in other word, this paper will be one of the causes to push the CGM run to lower redshift)</td></tr>
<tr><td>Subgrid physics</td><td>Same as the planned CGM run above.</td></tr>
<tr><td>Target redshift</td><td>z=0</td></tr>
<tr><td>Output strategy</td><td>See Section 4-(2) below.</td></tr>
<tr><td>Analysis tools</td><td>We may begin from metallicity analysis routines in agora-analysis-script used for our disk comparison paper (formerly Paper "4") and build upon it.</td></tr>
<tr><td>Obs. counterpart</td><td>e.g., Advanced LIGO constraints/predictions in the local Universe (to be determined)</td></tr>
</table>
• Chakrabarti et al. have started a discussion, and to plan test problems using isolated ICs.</td></tr>
<tr><td>OTHER LINKS</td><td>• WG IX Workspace (Characteristics of Cosmological Galaxies)
• WG XII Workspace (Interstellar Medium)</td></tr>
</table>

(4) Topics That Weren't Selected

(4-1) Suggested by the Steering Committee

• Reviving/Refocusing Paper "NoSubgrid" (formerly Paper "3") : cold flows and accretion with no subgrid physics, see Section 3 below. [suggested by Mayer]

• Morphology of star-forming galaxies at z~2 (prolate/oblate), compaction, main-sequence star formation, etc.

• However, don't be too pessimistic, and try to get as many groups who are running high-resolution zoom-in simulations to compare results and discuss which of these results agree best with observations and how we can improve the entire simulation effort. [suggested by Primack]

(4-2) Suggested by You

• Please feel free to submit a proposal for a new science project to utilize the AGORA platform and produce a science-oriented paper in the next 1-2 years. Add your short proposal here (see examples in 2-(2) above), or send it to the **Project coordinator**. Make sure to prepare a 1-minute pitch for your idea in the upcoming telecon.

3. Progress Report for Other Sub-projects (as of Apr. 2017)

We are making progress, chiefly, in three different areas below (numbered tentatively).

<table>
<tr><th></th><th>Paper "DM-Highres"
(previously known as
Paper "2")</th><th>Paper "NoSubgrid"
(previously known as
Paper "3")</th><th>Paper "Disk-II"
(previously known as
Paper "4b")</th></tr>
<tr><td>GOALS</td><td>• Run Proof-of-concept-DM (~1e11 Ms halo) at higher resolution</td><td>• Run Proof-of-concept-NOSUBGRID (~1e11 Ms halo)</td><td>• Compare simulation codes using common disk ICs</td></tr>
<tr><td>LEADERS</td><td>• Oliver Hahn (OCA, Nice)
• Christine Simpson (HITS)</td><td>• Nick Gnedin (Fermilab)
• Piero Madau (UCSC)
• Britton Smith (SDSC)</td><td>• Santi Roca-Fabrega (HUJI)</td></tr>
<tr><td>PARTICIPANTS</td>
<td>• Code representatives:
<table>
<tr><th>Code</th><th>Contacts</th></tr>
<tr><td>ART-I</td><td>Ceverino, Klypin, Primack</td></tr>
<tr><td>ART-II</td><td>Gnedin, Feldmann, Li, Semenov (Leitner)</td></tr>
<tr><td>CHANGA</td><td>Quinn, Wallace</td></tr>
<tr><td>ENZO</td><td>Kim, Hummels, Simpson, Wise</td></tr>
<tr><td>GADGET</td><td>Choi, Nagamine, Onorbe, (Rocha)</td></tr>
<tr><td>GADGET-AFS/CFS</td><td>(Hobbs)</td></tr>
<tr><td>GASOLINE</td><td>Madau, Shen, (Pillepich, Zolotov)</td></tr>
<tr><td></td><td></td></tr>
</table></td>
<td>• Code representatives:
<table>
<tr><th>Code</th><th>Contacts</th></tr>
<tr><td>ART-I</td><td>Ceverino, Klypin</td></tr>
<tr><td>ART-II</td><td>Gnedin, Feldmann, Leitner</td></tr>
<tr><td>CHANGA</td><td>Quinn</td></tr>
<tr><td>ENZO</td><td>Hummels, Kim, Simpson, Smith, Turk, Wise</td></tr>
<tr><td>GADGET</td><td>Choi, Nagamine</td></tr>
<tr><td>GADGET-SPHS</td><td>(Hobbs)</td></tr>
<tr><td>GASOLINE</td><td>Keller, Madau, Mayer, Shen, Wadsley</td></tr>
<tr><td>GEAR</td><td>Revaz</td></tr>
<tr><td>GIZMO</td><td>Lupi, Thompson</td></tr>
<tr><td>RAMSES</td><td>Agertz, Butler, Teyssier</td></tr>
</table>
(Grackle implementers; acknowledged in Paper "4")

• Working Groups engaged:</td>
<td>• Code representatives:
<table>
<tr><th>Code</th><th>Contacts</th></tr>
<tr><td>ART-I</td><td>Arraki, Ceverino, Klypin, Mandelker, Snyder</td></tr>
<tr><td>ART-II</td><td>Feldmann, Gnedin, Leitner, Li, Semenov</td></tr>
<tr><td>CHANGA</td><td>Quinn, Wallace</td></tr>
<tr><td>ENZO</td><td>Abel, Butler, Butsky, Forbes, Hummels, Kim, O'Shea, Simpson, Turk, Wise</td></tr>
<tr><td>GADGET</td><td>Choi, Nagamine, Chakrabarti, Shimizu, (Todoroki)</td></tr>
<tr><td>GADGET-SPHS</td><td>(Hobbs)</td></tr>
<tr><td>GASOLINE</td><td>Benincasa, Keller, Madau,</td></tr>
</table></td></tr>
</table>

<table>
<tr>
<td></td>
<td>
<table>
<tr><td>GCD+</td><td>Kawata</td></tr>
<tr><td>GIZMO-AFS</td><td>Garrison-Kimmel, Lupi, Hopkins, Onorbe</td></tr>
<tr><td>PKDGRAV2</td><td>(Guedes, Kuhlen)</td></tr>
<tr><td>RAMSES</td><td>Hahn, Roca-Fabrega, Teyssier</td></tr>
</table>
(Code leaders as of Aug. 2016; others interested: Dekel, Feldmann, Kawata, Pallottini, Robles)

• Working Groups engaged:
– III (common cosmo ICs)
– IV (common analysis)
– VI (dwarf galaxies)
</td>
<td>
– I (common physics)
– III (common cosmo ICs)
</td>
<td>
<table>
<tr><td></td><td>Mayer, Shen, Wadsley</td></tr>
<tr><td>GEAR</td><td>Revaz</td></tr>
<tr><td>GIZMO</td><td>Lupi, Hopkins, Thompson</td></tr>
<tr><td>RAMSES</td><td>Agertz, Butler, DeGraf, Teyssier, Nickerson</td></tr>
</table>
(Code leaders previously for Paper "4", as of Aug. 2016)

• Working Groups engaged:
– II (common disk ICs)
– IV (common analysis)
– V (subgrid physics)
</td>
</tr>
<tr>
<td>PROGRESS MADE OR BEING MADE</td>
<td>
• Please visit Paper's Workspace where Christine Simpson, Oliver Hahn and others have been updating the progress.

• We will plan to have 5-code comparison. As of Aug. 2016, 3+ participating groups have completed and submitted the production-level results (~100 pc resolution) to NERSC.

• Joel will check with Anatoly about ART-I, and with Peter about yt integration of Rockstar and consistent tree.

• Consistent tree - python interface has been built by Britton Smith (this link).
</td>
<td>
• Either via Britton's original interface or through Nick’s API, GRACKLE is fully implemented (equilibrium cooling version) in all 9 participating codes. It is now fully tested via disk comparison (previously known as Paper "4").

• Code groups such as ART-I and GIZMO have carried out cosmological simulations with various AGORA ICs, even including sophisticated subgrid physics.

• Ideally we will run at least some codes with GRACKLE's non-equilibrium cooling mode.
</td>
<td>
• Please visit Paper's Workspace where the results for disk comparison are compiled (including those presented in our first disk comparison paper - previously known as Paper "4"). The agora-analysis-script repository has been accordingly updated. 500 Myr snapshots to be publicly released soon.

• We will use the existing isolated runs and analysis pipeline to push the next paper, Paper "Disk-II" (previously known as Paper "4b").

• Analysis items that were not covered in our first disk comparison paper (previously known as Paper "4") will be considered first; see this link.
</td>
</tr>
<tr>
<td>OTHER LINKS</td>
<td>
• Please see the progress report in Paper's Workspace and/or in WG III's Workspace.

• MUSIC Google group
</td>
<td>
• GRACKLE Google group

• GRACKLE download & documentation

• GRACKLE method paper (Smith et al. 2017)

• AGORA API for GRACKLE by Nick Gnedin
</td>
<td>
• Please see the progress report in Paper's Workspace and/or in WG V's Workspace.

• yt-AGORA Google group
</td>
</tr>
</table>

4. Recaps

(1) AGORA SMBH Prescription (Accretion and Feedback):

- Please visit **this link**.

(2) Strategy to Store Simulation Outputs:

- Please visit **this link**.

(3) Your Own yt Installation on NERSC:

- Please visit **this link**.

(4) yt-Sunrise Pipeline for AGORA on NERSC:

- Please visit **this link**.

(5) Mini-workshop Opportunities:

- Mini-workshop for individual paper groups to write papers: KIPAC@Stanford is willing to provide funding opportunities for working group leaders to organize mini-workshops (~10 people). An example was the **yt-AGORA mini-workshop** at UCSC in Mar. 2014.

(6) Useful Links:

- Quick links to the important pages in AGORA Workspace, including internal policies: **here**
- Internal policy regarding the disclosure of AGORA simulation results: **here**
- Collection of softwares relevant to AGORA effort: **here**

5. Your Feedback

Let us hear anything AGORA, including the scientific and organizational effort currently being made. Voice your opinion during the web conference! Feel free to share your thoughts in the comments section below, or on the **mailing list**. You can also directly contact the **project coordinator** anytime!